Swarm Behaviour with Machine Learning

Daniel Sanchez
Eastern New Mexico University 2026
Daniel.R.Sanchez@enmu.edu

Introduction

This project aims to identify the most efficient and accurate supervised machine learning model for classifying high-dimensional swarm behavior data. Using the UCI Swarm Behaviour datasets, the project compares models across three binary classification tasks: aligned/not aligned, flocking/not flocking, and grouped/not grouped. Dimensionality reduction techniques, including PCA, were explored in an attempt to improve model performance and runtime. However, simpler models without dimensionality reduction were found to be more effective overall because the datasets were highly separable in their original feature space.

Methodology

Datasets:
This project used three files from the UCI Swarm Behaviour dataset: Aligned, Flocking, and Grouped. Each dataset represents a separate binary classification task where the Class column identifies whether the swarm behavior is present or absent. After cleaning, the Aligned dataset contained 24,016 rows and 2,401 columns, the Flocking dataset contained 24,015 rows and 2,401 columns, and the Grouped dataset contained 24,016 rows and 2,401 columns. Each dataset included 2,400 numerical predictor features and one binary target column.

Preprocessing:
All predictor variables represented numerical swarm-movement measurements, so one-hot encoding was not required. The datasets were inspected to verify that the feature columns were stored as numeric values, column names were cleaned, and missing or malformed values were checked. Only the Flocking dataset contained a missing/malformed row, so that row was removed before modeling..

Training and Testing:
Each dataset was split into an 80% training set and a 20% held-out test set. Stratified sampling was used because the datasets had different class distributions, especially Aligned and Grouped, which were more imbalanced than Flocking. This helped preserve the original class proportions in both the training and testing sets.

Scaling and Dimensionality Reduction:
Scale-sensitive models used StandardScaler inside the pipeline. PCA(10–50) and SelectKBest(200) were tested to see whether a smaller feature space could improve runtime while preserving accuracy. Although some PCA-based models reached the final eligibility threshold for near-perfect F1-score, they were not selected because their inference times were less efficient. The final results showed that dimensionality reduction was not necessary because the full original feature space produced the strongest and most efficient final model.

Model Training and Evaluation:
Initial untuned models were evaluated using cross-validation on the training data and showed that several algorithms could achieve near-perfect classification performance. So, light hyperparameter tuning was performed and compared against the baseline models. The project compared multiple pipelines. F1-score was used as the primary evaluation metric because some datasets had class imbalance. Since most models already classified the data extremely well, there was limited room to improve predictive performance. Therefore, prediction time became an important secondary metric. The final model was selected by pairing near-perfect F1-score with the most efficient average prediction time, then evaluated on the held-out test set.

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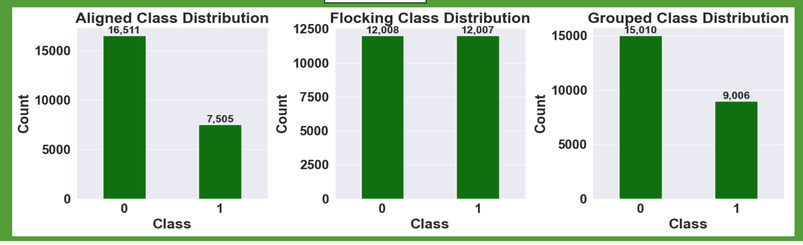
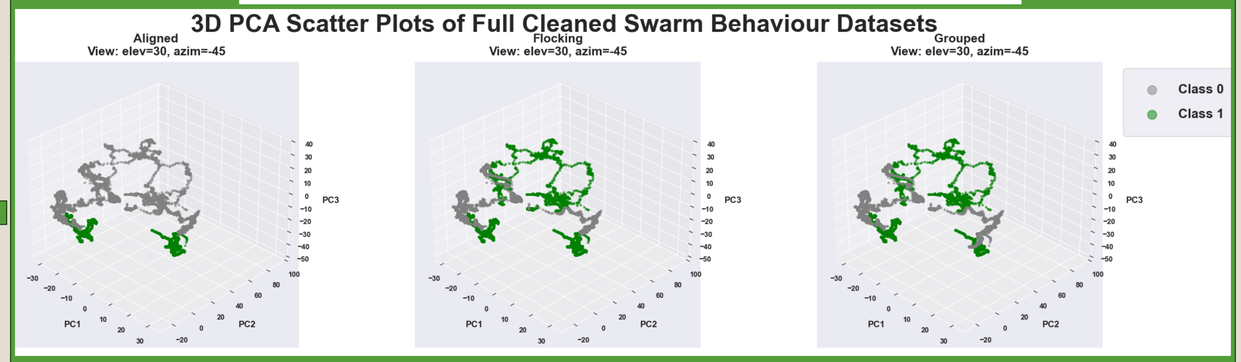
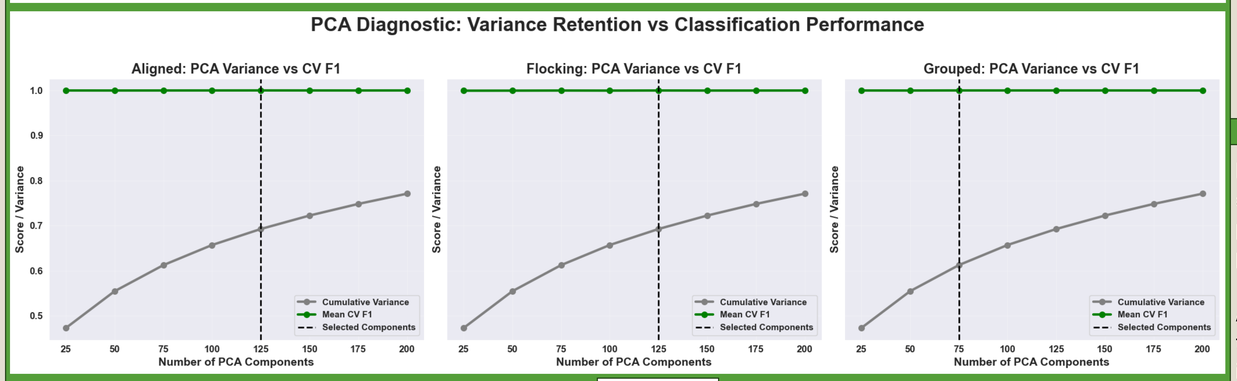
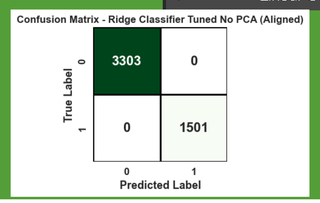


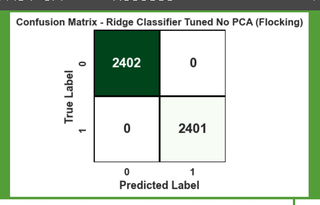


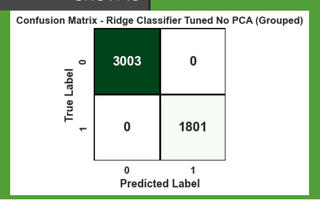


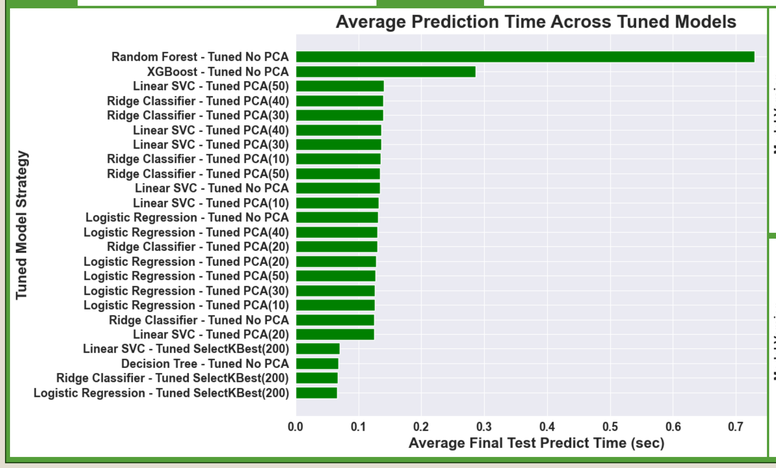
Results

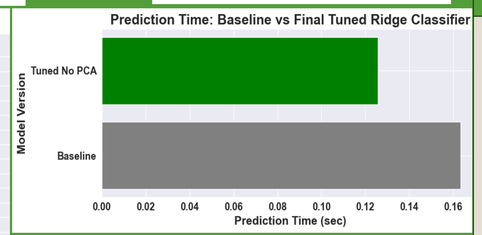
	Algorithm	Model Version	Avg Test F1	Avg Test ROC-AUC	Avg Prediction Time (sec)
0	Ridge Classifier	Tuned No PCA	1.000000	1.0	0.125760
1	Logistic Regression	Tuned PCA(30)	0.999907	1.0	0.126637
2	Logistic Regression	Tuned PCA(50)	1.000000	1.0	0.127542
3	Logistic Regression	Tuned No PCA	1.000000	1.0	0.131214
4	Linear SVC	Tuned No PCA	1.000000	1.0	0.134145

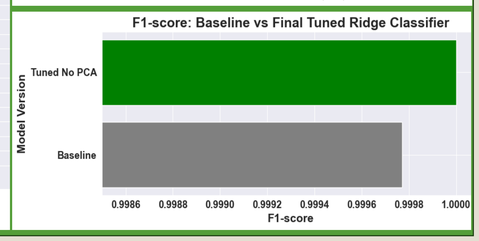












Conclusion

The results showed that the Swarm Behaviour datasets were highly separable using the original predictor features. Several final candidate models achieved 99.9%+ F1-score on held-out test evaluation, so final model selection considered both test F1-score and prediction time.

Ridge Classifier was selected as the final model because it achieved 99.9%+ test F1-score across all three datasets while also producing the fastest average prediction time among the eligible models. Logistic Regression with PCA(30) also performed very well, but the lightly tuned Ridge Classifier provided the best overall balance of classification performance and inference efficiency.

Although PCA-based dimensionality reduction did not improve performance enough to change the final model selection, several PCA models still reached the final eligibility threshold. This demonstrated that PCA could preserve strong classification performance, but the non-PCA Ridge Classifier remained the most efficient final choice.

Future Work

Future work could explore additional classification models and alternative modeling strategies to determine whether similar predictive performance can be achieved with even faster prediction times.

Additional feature-selection and dimensionality reduction techniques could also be tested to identify a smaller set of important swarm movement features while reducing model complexity.

Future experiments could compare different hyperparameter search methods, or more efficient optimization strategies, to reduce tuning time.

The model could also be tested on new swarm simulations to determine whether the same performance holds beyond the current UCI Swarm Behaviour datasets.

Random Forest and XGBoost will be investigated further because both performed extremely well at baseline, but their fit and prediction times were much higher than the final selected model.

References

Swarm Behaviour [Dataset]. (2020).
UCI Machine Learning Repository.
https://doi.org/10.24432/C5N02J